

# Battle of the Titans

The processor behemoths have started trading punches once again with Intel upping its game and AMD roughing it out every step of the way

## Business Performance and Office Productivity

The Pentium 4 2.4 GHz will wipe out the word 'sluggish' from your PC experience. Your PC will steadily hum along even with multiple applications working simultaneously. You will be able to switch between applications in the blink of an eye and response times will be instantaneous. All this is possible largely due to the Pentium 4's hefty 512 KB of full speed L2 cache. This coupled with truckloads of processing frequency makes a potent combination that beats the Athlon XP 2100+ by a margin of over 8 points in both Business Winstone and the I/O-heavy Content Creation benchmarks. The Intel Pentium 4 2.4 GHz takes user experience to a new level of smooth and unhindered computing bliss.

**Winner:** Pentium 4

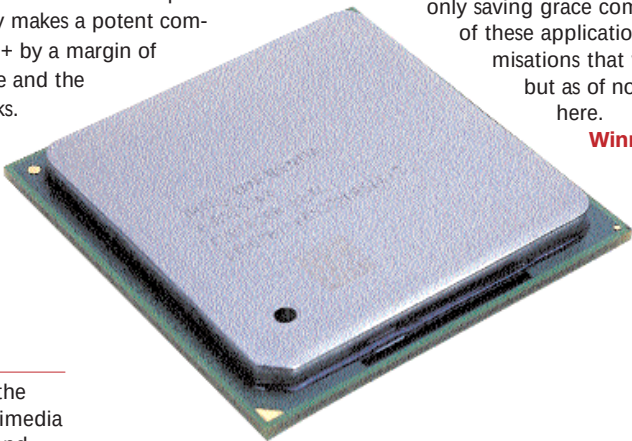
## Gaming and Multimedia

The raw speed of the Pentium 4 is all the more apparent when it comes to multimedia and games. SSE2 instructions and a bandwidth-quenching bus speed of 400 MHz ensure that the Pentium 4 can deal with almost anything you throw at it. Now your graphics card will have no excuses for churning out low frame rates. The Pentium 4, when paired with RDRAM, delivers unparalleled performance in our benchmarks—from the torturous 3DMark2001 SE to the legendary *Quake III Arena*—and beats the Athlon XP by a fair margin. We also bear witness to some of the highest recorded numbers in SiSoft Sandra's Multimedia benchmarks. To top it all, the processor hardly breaks a sweat running at a cool 42 degree Celsius!

**Winner:** Pentium 4

## How we Tested

We used the fastest available memory solutions officially endorsed by the manufacturer for each processor. The Pentium 4 was tested with 256 MB of PC800 RDRAM and an original Intel 850MD motherboard. The Athlon XP 2100+ was tested with 256 MB DDR SDRAM (PC2100) along with a VIA KT266A powered Gigabyte GA-7VTXE motherboard. The rest of the system remained common, consisting of nVidia GeForce3 64MB DDR graphics card (using Detonator 4 v21.83 drivers) and Seagate Barracuda 20 GB 7,200-rpm hard drive. We used Windows 2000 Professional SP2 with the latest patches and drivers.



## Price

This is one area where the Pentium 4 needs serious rationing, as Intel's fastest baby doesn't come cheap. A paltry sum of Rs 35,000 fetches you the undisputed bragging rights for owning the fastest processor on the planet! But you could buy an entire computer for the price of this processor alone. The hefty premium that the Pentium 4 demands is worsened when you factor in the price difference between DDR memory used by the AMD processor and RDRAM used by Intel.

**Winner:** Athlon XP

## Professional Applications

This is one area where the Pentium 4 fails to deliver the knock-out punch. Professional legacy applications like imaging and 3D software catch the Pentium 4 off-guard as it lacks the FPU muscle of the Athlon XP, and its frequency advantage can only help to a limited extent. Without SSE2 support in applications, the Pentium 4 lags considerably behind in our POV-Ray test. The only saving grace comes from the fact that future versions of these applications will include more Pentium 4 optimisations that will make up for its glaring deficiency, but as of now, the Pentium 4 needs crutches here.

**Winner:** Athlon XP

## Durability

The Pentium 4 has integrated thermal throttling, which automatically slows or shuts down the CPU if the core is experiencing very high temperatures. Each Pentium 4 processor has a metal cap (that functions as a heat spreader), so that the delicate CPU core is not exposed. The heat spreader also aids CPU life by increasing the surface area that's cooled. Additionally, Intel has made the heatsink install process completely idiot proof. Try as you might, you just cannot damage the core by incorrectly installing the heatsink.

**Winner:** Pentium 4

| Speed        |                                        |
|--------------|----------------------------------------|
| Category     | Benchmarks                             |
| Business     | Business Winstone 2001                 |
| Performance  | Content Creation 2001                  |
| Gaming       | 3DMark2001 SE                          |
| Performance  | Quake III (Normal) (FPS)               |
| Multimedia   | SiSoft Sandra Multimedia CPU (Integer) |
| Benchmarks   | SiSoft Sandra Multimedia FPU (SSE2)    |
| Professional | Video Encoding (in seconds)            |
| Benchmarks   | Photoshop 6.01 (in seconds)            |
|              | POV-Ray (Chess2) (in seconds)          |